

DIFFERENTIAL QUADRATURE METHOD IN THE BUCKLING ANALYSIS OF BEAMS AND COMPOSITE PLATES

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(Received 6 June 1990)

Abstract—This paper discusses the accuracy and convergence of the method of differential quadrature for solving a variety of differential equations with variable coefficients associated with plate and beam instability problems. The detailed solutions for buckling of rectangular, orthotropic and anisotropic, symmetric, angle-ply composite laminates under linearly varying uniaxial compression, uniaxial buckling of rectangular, isotropic plates with variable longitudinal thickness and lateral-torsional beam buckling under moment gradient are presented which highlight various computational aspects and, hopefully, contribute to a better understanding and judicious application of the method.

NOTATION

a	length of plate along x -axis
b	width of plate along y -axis
D	$= Et^3/12 (1 - \nu^2)$
D_m	$= Et_m^3/12 (1 - \nu^2)$
D_{ij}	bending stiffness matrix of a laminate
K_m	$= N_x a^2/\pi^2 D_m$
K_{av}	$= N_x a^2/\pi^2 \sqrt{(D_{11} D_{22})}$
k_{av}	$= N_x a^2/\pi^2 D$
M_u	uniform buckling moment of a simply supported I-beam
N	total number of DQ grid points
NP	total number of plies in a laminate
N_{av}	average uniaxial force/length at $x = a/2$
N_{min}	minimum compressive force/length ($x = 0$)
N_{max}	maximum compressive force/length ($x = a$)
N_x	uniaxial compressive force/length
N_{xy}	shear force resultant/length
r	N_{min}/N_{max}
t	plate thickness
t_0	plate thickness at $x = 0$
t_1	plate thickness at $x = a$
t_m	mean thickness $[(t_0 + t_1)/2]$
W	nondimensional plate deflection
X	nondimensional length ($= x/a$)
Y	nondimensional width ($= y/b$)
α	moment gradient
β	plate aspect ratio (a/b)
θ	fibre orientation in a lamina
λ	nondimensional buckling moment (M/M_u)

INTRODUCTION

In thin-walled, metal and composite structures, linear buckling is one of the most important design considerations. The mathematical description of stability problems of beams and plates of variable thickness or subjected to nonuniform loading involves differential equations with variable coefficients which cannot be solved exactly. Such problems are amenable to approximate solutions by variational (energy) methods. From practical considerations, it is felt that energy methods are not straightforward since they require

a priori selection of displacement functions satisfying, at least, the geometric boundary conditions, an uneasy task, especially in cases of mixed boundaries. Also, subsequent application of variational calculus often requires a sound knowledge of the principles of mechanics often beyond the grasp of many engineers. Numerical methods, such as finite elements or finite differences are less attractive due to excessive cost, storage requirements and data preparation. Besides the fact that energy methods, e.g. Ritz, Galerkin, etc., over the years, have proved to be powerful in dealing with a wide range of buckling problems, the principal interest lies in examining alternative techniques, at least from the theoretical point of view. This has motivated the search for an approximate technique for the direct solution of buckling problems without recourse to variational principles.

The paper proposes the method of the differential quadrature (DQ), as introduced by Bellman and Casti [1] and further elaborated by Civan and Sliepcevich [2], for solving directly the partial differential equation governing the problem with prescribed boundary conditions. This method suggests polynomial approximations of partial derivatives of a function. Jang *et al.* [3] have presented various applications of the DQM in structural mechanics. Earlier work [4] was focused on illustrating the scope and applicability of the method by means of a limited number of examples and the nature of the approximations was not discussed in great depth. The paper aims at clarifying some computational aspects, namely the effects of mesh refinement and mesh grading on the accuracy and convergence of the DQM, for a wide range of loading, boundaries and constitutive relations (orthotropic/anisotropic) such that its credibility for solving complex buckling problems can be assessed.

For this purpose, a representative example of the buckling of anisotropic composite plates under vari-

able compression and with various combinations of simple and clamped boundary conditions is considered. A detailed comparison of DQM solutions against standard results obtained by the reliable Ritz method [5], which assures a monotonically converging symmetric eigenvalue problem, allows a comprehensive assessment of this relatively new method. Also, buckling of variable thickness isotropic plates, and beams under nonuniform bending, are studied by DQM which broadens experience with the method. To our knowledge, this is the first time that the method has been applied and examined extensively in plate instability problems.

METHOD OF DIFFERENTIAL QUADRATURE

In the method of differential quadrature, partial space derivatives of a function appearing in a differential equation are approximated by means of a polynomial expressed as the weighted linear sum of the function values at a preselected grid of discrete points. For example, the first partial derivative of a function, $f(x)$, at the i th discrete point is approximated by

$$\frac{\partial f(x_i)}{\partial x} \cong \sum_{j=1}^N A_{ij} f(x_j), \quad i = 1, 2, \dots, N, \quad (1)$$

where, x_i are the set of discrete points in the x -direction and A_{ij} are the associated weighting coefficients which can be derived by assuming the function, $f(x)$, of the following polynomial form

$$f(x_i) = x_i^{k-1}, \quad k = 1, 2, \dots, N. \quad (2)$$

From (1) and (2), one can write

$$\sum_{j=1}^N A_{ij} f(x_j) = (k-1) x_i^{k-2}, \quad i, k = 1, 2, \dots, N. \quad (3)$$

This represents N sets of N linear, simultaneous algebraic equations which have a unique solution for the weighting coefficients, A_{ij} . The weighting coefficient for higher order partial derivatives can be obtained by an alternative technique of utilizing individual quadratures [2] which, in essence, implies that, for example, the second-order derivative can be written as

$$\left(\frac{\partial^2 f}{\partial x^2}\right)_i = \sum_{j=1}^N B_{ik} f(x_k), \quad (4)$$

where B_{ik} is obtained by simple matrix multiplication, $[B] = [A] \times [A]$. The weighting coefficient for third and fourth-order derivatives, C_{ij} and H_{ij} , respectively, can be derived in a similar way, $[C] = [A] \times [B]$ and $[H] = [B] \times [B]$. A subroutine is written for computing weighting coefficients, A_{ij} , B_{ij} , C_{ij} , and H_{ij} , for a

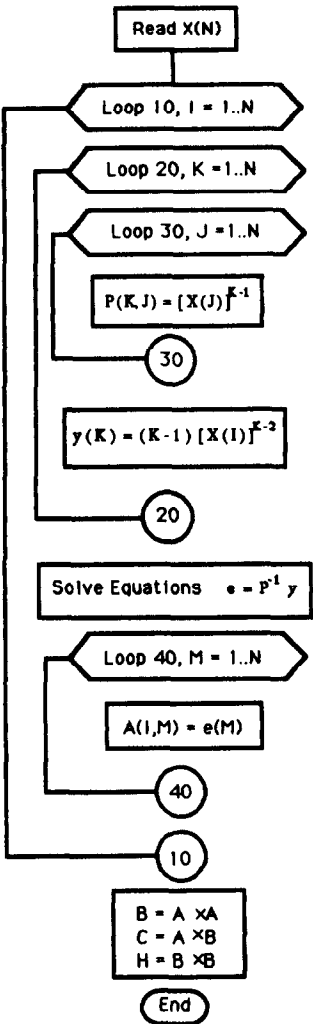


Fig. 1. Flowchart for weighting coefficients.

preselected grid of points, the flowchart for which is given in Fig. 1.

Presently, grids consisting of 6–9 points are employed for solving beam and plate buckling problems which are reported in Table 1. The effect of the relative spacing of points is studied by two grids of 9 points, denoted as 9(I) and 9(II). In all plate buckling problems, grids in X and Y -directions are

Table 1. Coordinates of points for DQ grids

No.	x/a				
	$N = 6$	$N = 7$	$N = 8$	$N = 9(I)$	$N = 9(II)$
1	0	0	0	0	0
2	0.0001	0.0001	0.0001	0.0001	0.0001
3	0.333	0.25	0.2	0.166	0.14
4	0.67	0.5	0.4	0.332	0.4
5	0.9999	0.75	0.6	0.498	0.5
6	1	0.9999	0.8	0.664	0.6
7		1	0.9999	0.83	0.86
8			1	0.9999	0.9999
9				1	1

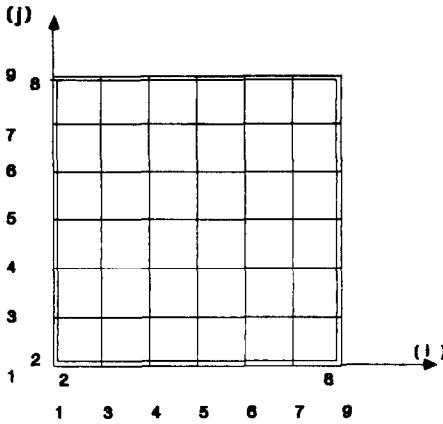


Fig. 2. Mesh for DQ approximation.

chosen to be the same. A typical DQ mesh for plate problems is shown in Fig. 2.

APPLICATIONS

In this section, three buckling problems, namely a laminated composite plate under variable compression, a variable thickness plate under constant uniaxial compression and the lateral-torsional buckling of an I-beam under unequal bending moment, all represented by differential equations with variable coefficients, are analysed by the differential quadrature method (DQM) and results are compared with available standard solutions.

1. Buckling of composite plates under variable compression

The differential equation of equilibrium of a rectangular anisotropic plate [6] under uniaxial compression, N_x , and shear force, N_{xy}

$$D_{11}w_{,xxxx} + 2(D_{12} + 2D_{33})w_{,xxyy} + D_{22}w_{,yyyy} + 4D_{13}w_{,xxyy} + 4D_{23}w_{,xyyy} = N_x w_{,xx} + 2N_{xy}w_{,xy} \quad (5)$$

combined with the boundary conditions, defines the plate buckling problem of current interest. The subscripts preceded by a comma denote differentiation with respect to the corresponding coordinates. Here, D_{ij} , $i, j = 1, \dots, 3$ are the bending stiffness of the laminated plate [7], w is the out-of-plane deflection and $\beta = a/b$ is the plate aspect ratio. In orthotropic approximations, $D_{13} = D_{23} = 0$, while $D_{11} = D_{22}$ for isotropic plates. It is assumed that the plate is subjected to linearly varying compressive forces per unit length $N_{\min}$ at $x = 0$ and $N_{\max}$ at $x = a$ (Fig. 3) such that compressive stress at any section x can be written as

$$N_x = 2N_{av}(C_1 X + C_2) \quad \text{where} \quad N_{av} = N_{\max} \frac{(1+r)}{2} \quad (6)$$

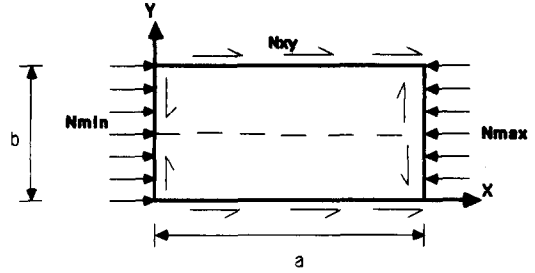


Fig. 3. Plate buckling under compression gradient.

Here, $C_1 = (1-r)/(1+r)$, $C_2 = r/(1+r)$ and $r = N_{\min}/N_{\max}$. N_{av} is the average compressive stress and r is the ratio of minimum to maximum compressive stress. The conditions of equilibrium of mid-plane stresses

$$\frac{\partial N_x}{\partial x} + \frac{\partial N_{xy}}{\partial y} = 0 \quad \text{and} \quad \frac{\partial N_y}{\partial y} + \frac{\partial N_{xy}}{\partial x} = 0 \quad (7)$$

provide the following expression for shear stress distribution [8] in the prebuckling state

$$N_{xy} = -\frac{N_{av}}{\beta} C_1 (2Y - 1). \quad (8)$$

$X = x/a$ and $Y = y/b$ are nondimensional coordinates. The boundary conditions may be defined, for simple supports, as

$$w = 0 \quad \text{at} \quad x = 0, a \quad \text{and} \quad y = 0, b$$

$$M_x = D_{11}w_{,xx} + D_{12}w_{,yy} + 2D_{13}w_{,xy} = 0 \quad \text{at} \quad x = 0, a \quad (9)$$

$$M_y = D_{12}w_{,xx} + D_{22}w_{,yy} + 2D_{23}w_{,xy} = 0 \quad \text{at} \quad y = 0, b$$

and, for clamped supports, as

$$w = 0 \quad \text{at} \quad x = 0, a \quad \text{and} \quad y = 0, b$$

$$w_{,x} = 0 \quad \text{at} \quad x = 0, a \quad \text{and} \quad w_{,y} = 0 \quad \text{at} \quad y = 0, b. \quad (10)$$

The edge conditions are denoted by the letter S for simple and C for clamped supports along the four edges in the following order, $x = 0$, $x = a$, $y = 0$ and $y = b$. Thus, support conditions denoted by SSCS means that the $x = 0$, a and $y = b$ edges are simply supported and the $y = 0$ edge is clamped.

Using the nondimensional coordinates, X , Y , and W , the differential equation of equilibrium, (5),

and for clamped supports

$$W_{1j} = W_{Nj} = W_{i1} = W_{iN} = 0, \quad i, j = 1, \dots, N$$

$$\sum_{k=1}^N A_{ik} W_{kj} = 0, \\ i = 2, (N-1) \quad \text{and} \quad j, k = 2, \dots, (N-1)$$

$$\sum_{k=1}^N A_{jk} W_{ik} = 0, \\ j = 2, (N-1) \quad \text{and} \quad i, k = 2, \dots, (N-1). \quad (13)$$

Mixed boundary conditions can easily be accommodated by combining (12) and (13). The method can be extended to include rotational boundary restraints as well. In the formulation, zero edge displacement conditions are obviously applied at $X, Y = 0, 1$ but zero moment or slope conditions, as the case may be, are applied at grid points very close to the plate boundaries. A computer program is developed which, for an assumed grid of points in the X and Y -directions, computes weighting coefficients for partial derivatives, generates the bending stiffness matrix from specified lamination parameters and finally solves the generalized eigenvalue problem

defined by (11) with appropriate B.C. using the QZ algorithm.

The results are reported for a plate with four different combinations of simple and clamped boundaries under three loading conditions corresponding to the compression gradient, $r = 1, 0.5, -0.5$. Three grids of 8, 9(I), and 9(II) are employed. In the 8 and 9(I) grid, points in the middle of the plate are uniformly spaced; the 9(II) grid has nonuniformly spaced points which were chosen after many trials. The present analysis is limited to mid-plane symmetric laminates which are more popular for practical use compared with unsymmetric laminates where bending-stretching coupling results in undesirable twisting/warping due to normal changes in temperature, humidity, etc. Nondimensional average buckling stresses, $K_m = N_{av} a^2 / \sqrt{(D_{11} D_{22})}$, for square, symmetric angle-ply laminates with fibre angle $\theta = 0-90^\circ$ in steps of 15° are presented in Figs 4-9 for B.C. SSSS, SSSC, SSCS, CCCC, CCSS, and CCCS, respectively. In a symmetric angle-ply laminate, the fibre angle is θ for even-numbered and $-\theta$ for odd-numbered lamina and the total number of laminae, NP, is odd, presently, NP = 3. The bending stiffness matrix, D_{ij} , of such laminates can be written in terms of generic orthotropic constants and fiber angle, θ [9]. The material considered herein is graphite-epoxy composites having longitudinal

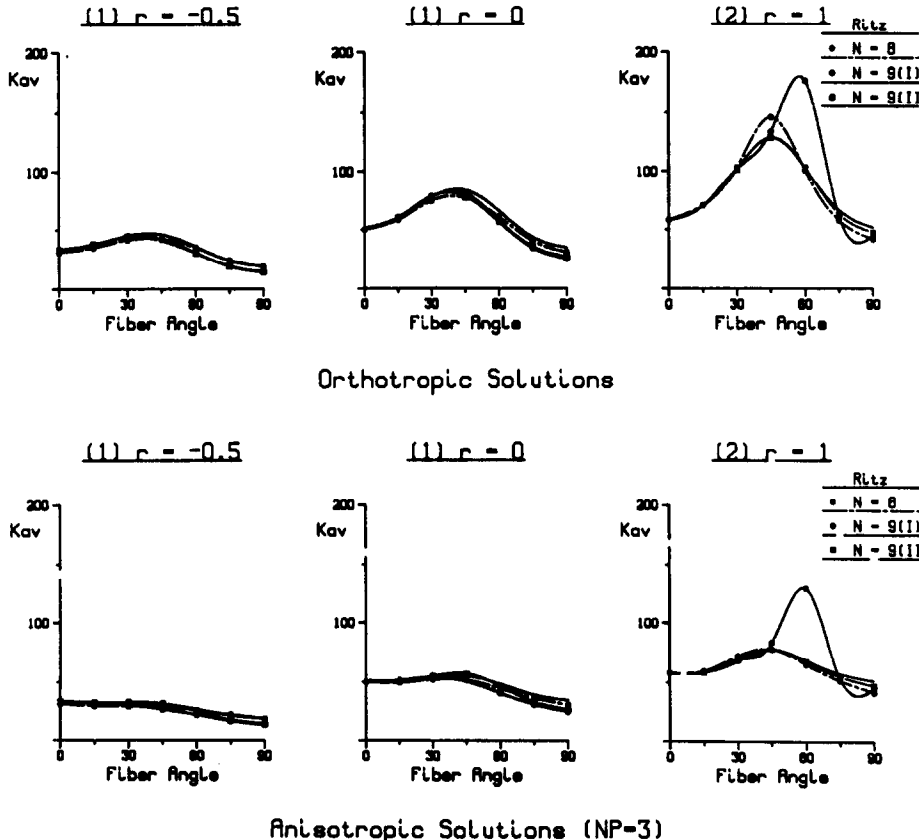


Fig. 5. Buckling of SSCC plates.

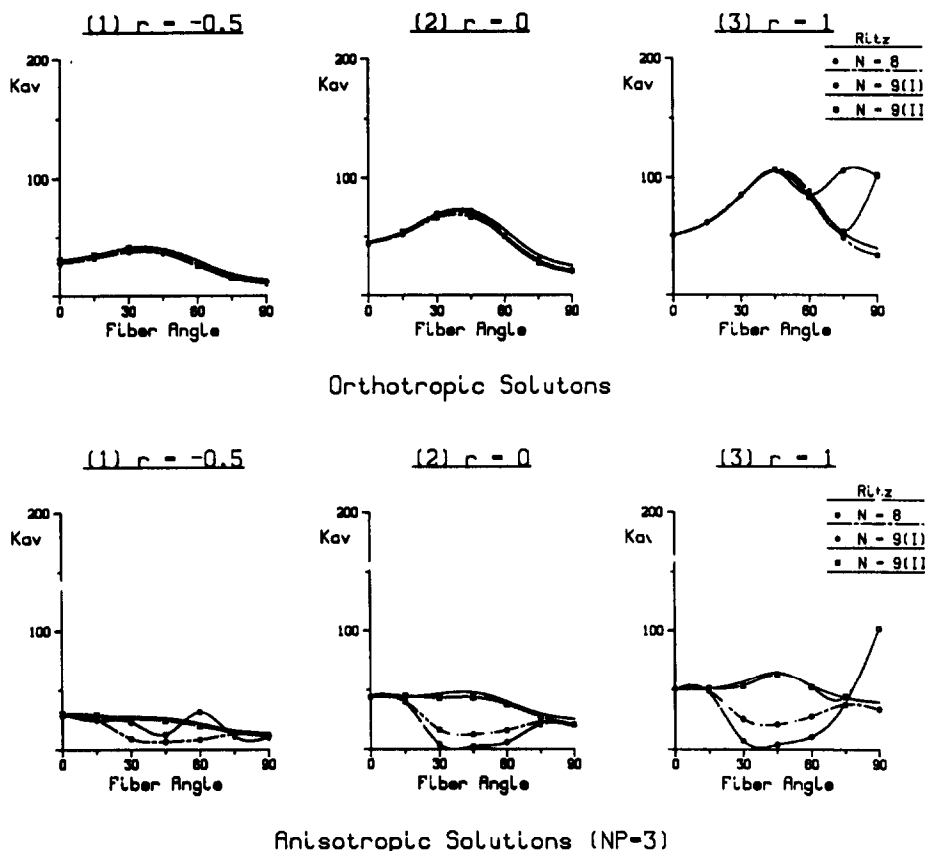


Fig. 6. Buckling of SSCS plates.

modulus, $E_{11} = 20.5 \times 10^6$ psi, transverse modulus, $E_{22} = 1.4 \times 10^6$ psi, shear modulus, $G = 0.59 \times 10^6$ psi, and major Poisson's ratio, $\nu_{12} = 0.26$. The DQM solutions are compared with more accurate and reliable results obtained by the Ritz method in conjunction with orthogonal polynomials (six terms), generated by a Gram-Schmidt process, the details of which can be found elsewhere [5].

Discussion. The results obtained for six combinations of simple and clamped end conditions for orthotropic ($D_{13} = D_{23} = 0$) as well as anisotropic plates (NP=3) are presented in Figs 4-9 together with the results obtained by the Ritz method. This particular example is very interesting as it reveals the importance of the material orthotropy/anisotropy, loading gradient, fiber angles and boundary conditions on numerical performance as the number of grid points and their relative spacing is varied. The term 'numerical performance' refers to the numerical accuracy and convergence of DQ approximations against some standard benchmark solutions. The variation in results obtained by 9(I) and 9(II) grids is treated as a measure of sensitivity to the relative spacing of grid points. Firstly, nondimensional buckling coefficients, $k_w = N_x a^2 / \pi^2 D$, for square isotropic SSSS plates under various compression gradients are compared (Table 2) with the standard solutions [8] obtained using the Rayleigh-Ritz method.

The DQM solutions with the 9(II) grid compares very well.

In cases of anisotropic, composite plates with SSSS and SSCS edge conditions (Figs 4 and 6), the numerical performance of 8 and 9(I) grids is very poor, especially in the range of fibre angles, $30 \leq \theta \leq 60$, which questions their reliability. At the same time, in the orthotropic approximation, remarkable accuracy is attained by employing 8 and 9(I) grids for the complete range of fibre angles with the exception of the 9(I) grid for the SSCS plate at $\theta = 60$ and $r = 1$. Contrary to this, in all the cases, DQM solutions adopting the 9(II) grid are in close agreement with those obtained by the Ritz method [5]. It is to be noted that laminates with fibre angle, $30 \leq \theta \leq 60$, exhibit pronounced anisotropy and the DQ formulation for anisotropic plates with simply supported loaded edges is highly sensitive to the relative spacing of grid points apparently caused by this structural anisotropy. It is believed that the anisotropic terms, D_{13} , D_{23} , complicate the buckling mode shape, which cannot be encapsulated by 8 and 9(I) grids and, thus, a judicious spacing of points becomes a prerequisite for reliable numerical results. These anisotropic coupling terms are also involved in the moment boundary condition equations which further complicates the convergence behaviour. The buckling behaviour of the SSCC plate is somewhat different as the DQM

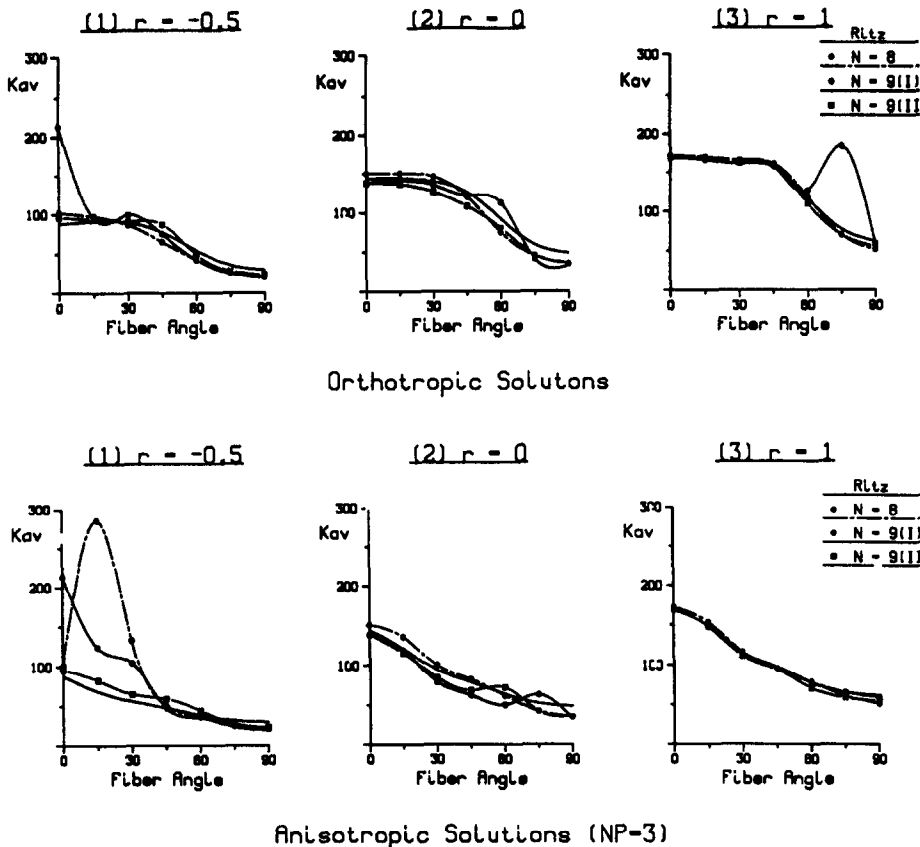


Fig. 7. Buckling of CCCC plates.

solutions obtained by all the three grids are reasonably accurate with the exception of 9(I) grid under load gradient, $r = 1$.

For plates with clamped loaded edges, the loading gradient rather than anisotropy has significant impact on the numerical performance. It is evident from the fact that, for all the three cases of CCCC, CCSS, and CCCS plates (Figs 7–9) under compression gradient, $r = -0.5$, and fibre angle, $0 < \theta < 30$, the performance of DQM is unsatisfactory for both the orthotropic and anisotropic laminates. It should be noted that anisotropic coupling terms are significantly reduced when the fibre angle varies within the range $0 \leq \theta \leq 30$. In general, the performance is reasonable when the loading is more uniform, e.g. $r = 0, 1$ which reinforces an earlier conjecture. Occasionally, these grids lead to inaccurate results, e.g. CCCC orthotropic plates, $r = 1$, with $N = 8$, CCCS anisotropic plates, $r = 0$, with $N = 8$ and 9(I). It must be emphasized that clamped boundary conditions of zero displacement and slope do not involve anisotropic terms and, therefore, solutions become relatively insensitive to the anisotropy in comparison with the loading gradient.

It has been reported that the Ritz method is computationally expensive over the DQ method [5]. The DQM with NP grid points leads to a generalized

eigenvalue problem involving square matrices of dimensions $(N-2)^2 \times (N-2)^2$ whereas the Ritz method with NT terms in orthogonal polynomial sequences leads to a problem of size $NT^2 \times NT^2$. Thus, with nine grid points in the DQM and a six-term expansion in the Ritz method, the problem size would be 49×49 and 36×36 , respectively. Despite the smaller eigenvalue problem, the computational effort required for evaluating energy integrals in the Ritz method is far greater than calculating the weighting coefficients in the DQM which makes the Ritz method, overall, significantly expensive.

2. Buckling of variable thickness plates

The differential equation of equilibrium of isotropic plates, with variable rigidity due to linear taper along the plate length and subjected to uniaxial compression, is given by [10]

$$\begin{aligned}
 & D[W_{,xxxx} + 2\beta^2 W_{,xxyy} + \beta^4 W_{,yyyy}] \\
 & + 2 \frac{\partial D}{\partial X} [W_{,xxx} + \beta^2 W_{,xyy}] \\
 & + \frac{\partial^2 D}{\partial X^2} [W_{,xx} + \nu \beta^2 W_{,yy}] = a^2 N_x W_{,xx}. \quad (14)
 \end{aligned}$$

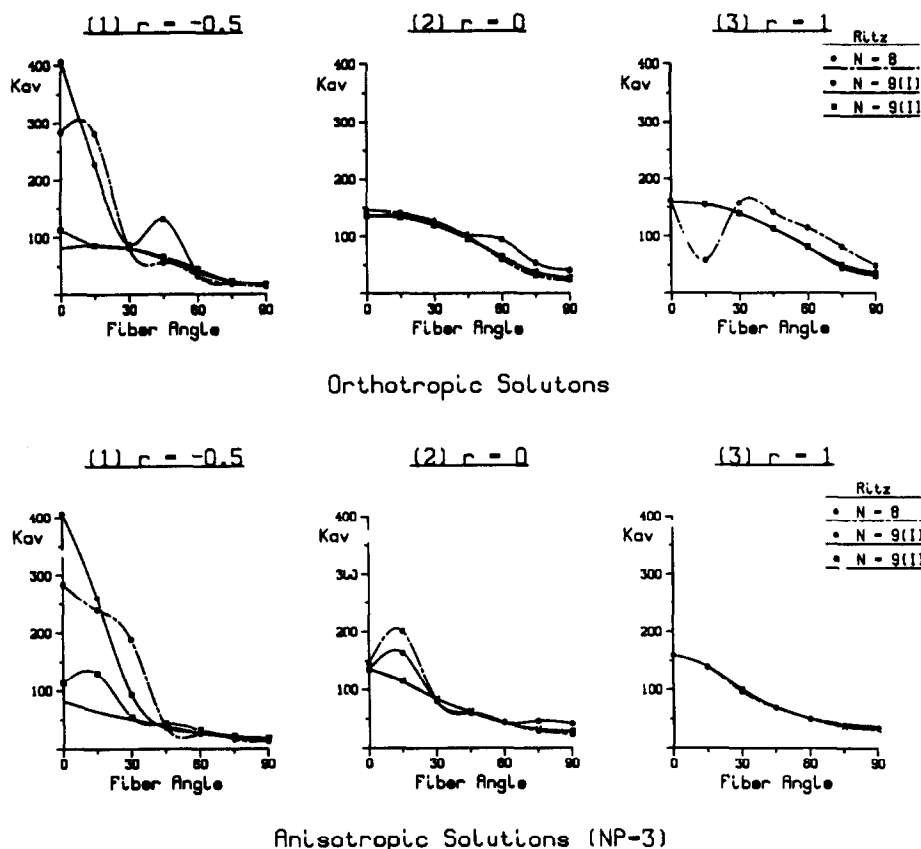


Fig. 8. Buckling of CCSS plates.

Here, it is assumed that the thickness varies linearly in the loading direction, X , from t_0 ($X=0$) to t_1 ($X=1$) as shown in Fig. 10. The thickness function can be written in terms of mean thickness, $t_m = (t_1 + t_0)/2$, at $X=0.5$, as

$$t(X) = t_m[1 + c(X - 0.5)] \quad \text{where} \quad c = \frac{2(t_1/t_0 - 1)}{(t_1/t_0 + 1)} \quad (15)$$

such that bending rigidity at any section is given by

$$D(x) = D_m f^3(X) \quad \text{where}$$

$$f(X) = [1 + c(X - 0.5)] \quad \text{and}$$

$$D_m = Et_m^3/12(1 - \nu^2). \quad (16)$$

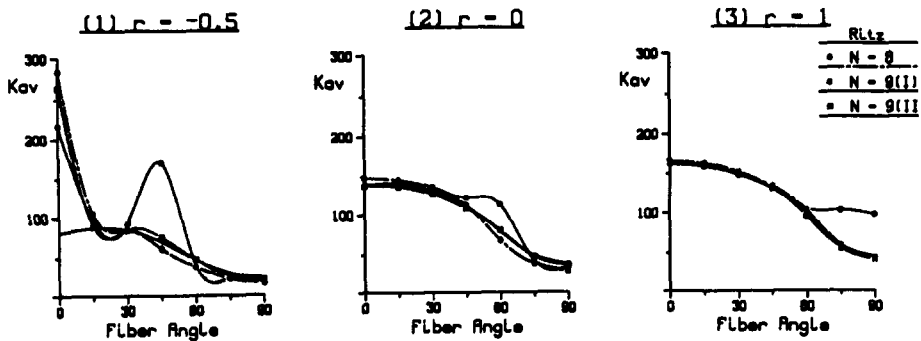
Now, substituting for bending rigidity and its derivatives in accordance with (16) into (14) and replacing the displacement derivatives by quadrature approximations, we obtain the following equations

$$f^3(X) \left[\sum_{k=1}^N H_{ik} W_{kj} + 2\beta^2 \sum_{m=1}^N B_{jm} \sum_{k=1}^N B_{ik} W_{km} + \beta^4 \sum_{k=1}^N D_{jk} W_{ik} \right]$$

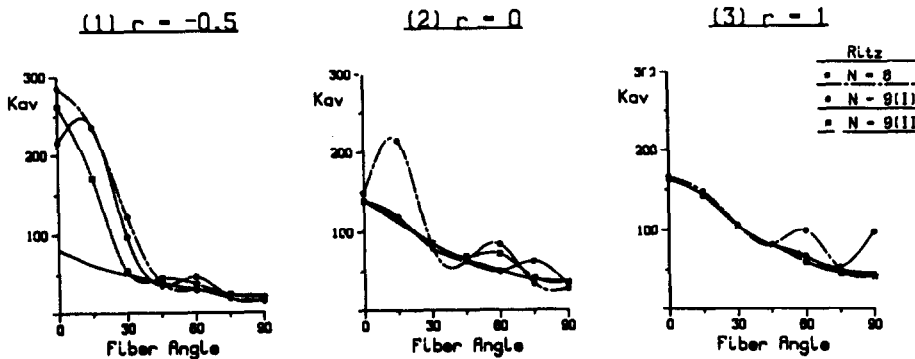
$$+ 6cf^2(X) \left[\sum_{k=1}^N C_{ik} W_{kj} + \beta^2 \sum_{m=1}^N B_{jm} \sum_{k=1}^N A_{ik} W_{km} \right] + 6c^2f(X) \left[\sum_{k=1}^N B_{ik} W_{kj} + \nu\beta^2 \sum_{k=1}^N B_{jk} W_{ik} \right] = K \left[\sum_{k=1}^N B_{ik} W_{kj} \right], \quad i, j = 3, \dots, (N-2). \quad (17)$$

Here, $K = N_x a^2/D_m$.

Wittrick and Ellen [11] reported solutions for SSSS and CCSS plates using the Rayleigh-Ritz method which are compared with DQ results in Tables 3 and 4 for square plates. In Table 3, lower and upper bounds obtained by Chehil and Dua [10] are also enumerated. For moderately tapered SSSS plates, i.e. $t_1/t_0 \leq 1.5$, results obtained by 9(I) and 9(II) grids are quite close indicating little sensitivity to the relative spacing of grid points. In the case of SSSS plates, the results obtained by the 9(II) grid provide a close lower bound and, for CCSS plates, a grid of $N=8$, provides a close upper bound to the classical solutions [11]. Buckling loads are reported in nondimensional form, $K_m = N_x a^2/\pi^2 D_m$.



Orthotropic Solutions



Anisotropic Solutions (NP-3)

Fig. 9. Buckling of CCCS plates.

3. Lateral-torsional buckling of beams

The differential equation governing lateral-torsional buckling of a beam of doubly symmetric cross-section and subjected to linearly varying bending moment can be written in terms of angle of twist, ϕ , as follows [12]:

$$EC_w \phi_{,xxxx} - GJL^2 \phi_{,xx} = \frac{M^2(X)L^4}{EI_y} \phi, \tag{18}$$

where I_y , C_w , and J denote minor axis moment of inertia and the torsional and warping rigidity, respectively, of the beam cross-section and L is the beam length. The linear variation of bending moment is given as

$$M(X) = M[\alpha + (1 - \alpha)X], \tag{19}$$

Table 2. Buckling coefficient, k_m , for isotropic, square plates under variable compression

No.	r	Ref. [8]	DQM		
			$N = 8$	$N = 9(I)$	$N = 9(II)$
1	-0.33	2.43	2.2	2.38	2.43
2	-0.2	2.77	2.55	2.75	2.77
3	0	3.2	3.02	3.21	3.21
4	0.2	3.53	3.41	3.56	3.55
5	0.5	3.85	3.81	3.87	3.86
6	1	4	4.01	4	4

where, α is the moment gradient as shown in Fig. 11. Boundary conditions are given as,

simple support (SS)

$$\phi(0) = \phi(1) = \phi_{,xx}(0) = \phi_{,xx}(1) = 0$$

warping fixed (WF)

$$\phi(0) = \phi(1) = \phi_{,x}(0) = \phi_{,x}(1) = 0.$$

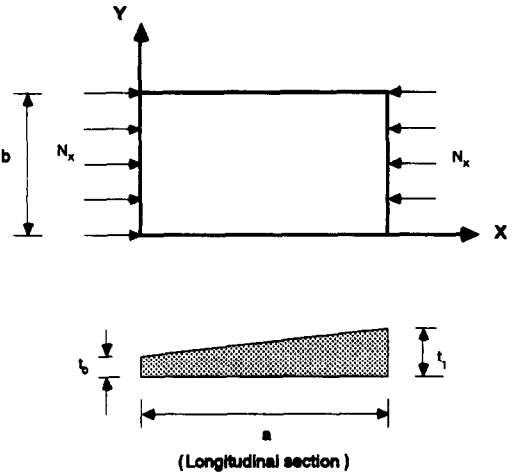


Fig. 10. Variable thickness plate under uniaxial compression.

Table 3. Buckling coefficient, K_m , for square, variable thickness SSSS plates

No.	t_1/t_0	Ref. [11]	Ref. [10]		DQM		
			L.B.	U.B.	$N = 8$	$N = 9(I)$	$N = 9(II)$
1	1.125	3.966	3.966	3.996	3.897	3.92	3.917
2	1.25	3.882	3.878	3.886	3.761	3.843	3.837
3	1.5	3.638	3.604	3.687	3.481	3.573	3.592
4	1.75	3.364	3.264	3.549	3.297	3.237	3.299
5	2	3.1	2.9	3.537	3.234	3.291	3.005

Table 4. Buckling coefficient, K_m , for square, variable thickness CCSS plates

No.	t_1/t_0	Ref. [11]	DQM		
			$N = 8$	$N = 9(I)$	$N = 9(II)$
1	1.125	6.678	6.71	6.619	6.619
2	1.25	6.51	6.575	6.558	6.47
3	1.5	6.036	6.152	6.317	6.147
4	1.75	5.466	5.653	5.924	5.872
5	2	4.912	5.178	5.442	5.636

The differential quadrature approximation of this equation is written as

$$EC_w \sum_{k=1}^N H_{ik} \phi_k - GJL^2 \sum_{k=1}^N B_{ik} \phi_k$$
$$= \frac{M^2 L^4}{EI_y} [\alpha + (1 - \alpha) X_i]^2 \phi_i, \quad (20)$$

where, $i, j = 3, \dots, (N - 2)$. This eigenvalue problem is solved for simple and warping fixed supports and their combination and results are compared (Table 5) with standard finite element solutions [13]. The numerical results are reported for a typical I-beam cross-section W 460 $\times$ 82 (Canadian designation) and $L = 8.633$ m. The properties of the beam section are $I_y = 18.6 \times 10^6 \text{ mm}^4$, $C_w = 0.918 \times 10^{12} \text{ mm}^6$, and $J = 0.691 \times 10^6 \text{ mm}^4$. The elastic and shear moduli are $E = 200 \text{ GPa}$ and $G = 77 \text{ GPa}$, respectively. The buckling moment is reported in nondimensional form, $\lambda = M/M_{cr}$, where M_{cr} is the uniform buckling moment of a simply supported I-beam. A grid of N points leads to an eigenvalue problem of size $(N - 2) \times (N - 2)$.

DQM solutions provide fairly good estimate of lateral-torsional buckling moment for a beam bending in single curvature, i.e. $\beta \leq 0$. In all cases, results

obtained by grid $N = 8$ with resulting eigenvalue problem of size 6×6 are comparable with those obtained by six finite elements with a problem size of 20×20 . The beam finite elements based on cubic polynomials as shape functions have four degrees of freedom at each node.

CONCLUDING REMARKS

It is clear that the quadrature method (DQM) offers a very compact procedure for buckling analysis which appears to be more attractive than the classical Ritz method provided that reasonable accuracy can be obtained. In this method, boundary conditions of zero moment, even in the case of anisotropic plates, can be easily satisfied on a point-wise basis which could not be achieved using orthogonal polynomials in the Ritz formulation [5]. Experience shows that the DQ method does not result in monotonically converging results as the mesh is further refined. Also, results were found to be highly sensitive to the number and relative spacing of grid points. The DQ method is conceptually a polynomial fit to the derivative of a function and, hence, subjected to its limitations. The accuracy of the approximation increases as the number of grid points or the order of the polynomial increases. Beyond a certain point, numerical ill-conditioning can result which introduces gross errors. In the formulation, zero edge displacement conditions are obviously applied at $X, Y = 0, 1$ but zero moment or slope conditions, as the case may be, are applied at points very close to the plate boundaries which further contribute to numerical instability/errors in the higher order approximation. The present study indicates that numerical performance of the DQM depends on various problem parameters. For example, anisotropic plates with

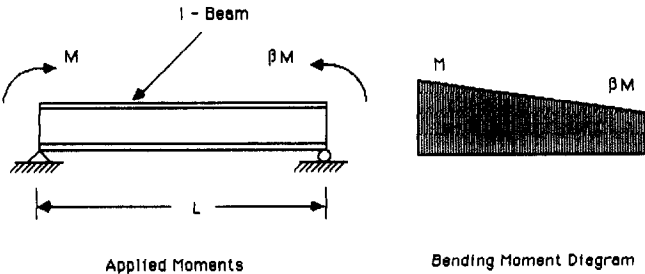


Fig. 11. Beam buckling under unequal end moments.

Table 5. Buckling coefficient, λ , for I-beams under nonuniform bending

No.	Supports	β	F.E.M.	D.Q.M		
				$N = 6$	$N = 7$	$N = 8$
1	S-S	1	1	0.984	1.001	1
		0	1.813	1.804	1.787	1.8106
		-1	2.604	2.92	2.447	2.543
2	S-WF	1	1.235	1.237	1.247	1.25
		0	2.116	2.045	2.0253	2.077
		-1	3.466	3.657	3.161	3.192
3	WF-WF	1	1.567	1.584	1.578	1.572
		0	2.986	2.873	2.907	2.926
		-1	4.975	4.7	4.666	4.896

simply supported loaded edges are very sensitive to the mesh grading, i.e. relative spacing of points, which increases with load gradient but disappears in orthotropic approximations. In cases of clamped loaded edges, load gradient rather than anisotropy dictates the accuracy of the approximation. Presently, a nonuniform mesh of 9 points, 9(II), consistently exhibits stable behaviour with reasonable accuracy in all plate problems. In this grid, points are closely spaced in the central zone of the plate which may suggest a tentative criterion for selecting them, that is, more points should be selected in the expected region of higher deformation (strain energy). This particular grid has been adopted after many trials. Experience shows that compactness and computational economy of the DQ model is praiseworthy. The accuracy of DQM, which is assessed in detail against benchmark solutions, is found to be satisfactory provided that a suitable grid is selected.

Acknowledgements—The authors are grateful to NSERC of Canada for support of this work under grant A-1582 to the principal author (A.N.S.) and to the Province of Ontario,

Canada, for support of the second author (M.D.P.) under an Ontario Graduate Scholarship.

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